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Candidate Rotation Periods for 34 Main-belt Asteroids from Gaia DR3 Sparse Photometry

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ABSTRACT

I searched Gaia DR3 sparse photometry of 598 well-observed numbered asteroids for rotation periods, fitting at each trial frequency a linear model with season offsets, a linear phase-angle term and a two-harmonic Fourier series. For confident results whose period was not used to tune the method, 96 of 122 (79%) agree within 3% with periods published by others; accuracy is low above 50 h, where the method usually returns half of the true period. I list 34 asteroids whose confident period was absent from all major period compilations, as targets for confirmation; a second analysis of the same data with a refined method agrees within 0.2% for each of them (seven candidates it contradicted were dropped).

DATA

I ranked the numbered asteroids in the Gaia DR3 `sso_source` table (Gaia Collaboration et al. 2023) by their number of observations and kept the 6,000 best observed. Going down this list, I took the first 300 that have a rotation period of 2–100 h in JPL SBDB, which follows the October 2023 LCDB (Warner et al. 2009) (*validation set*), and the first 300 that have no JPL period, of which 298 had enough data (*target set*).

Gaia reports one G magnitude per CCD crossing in `sso_observation`. For each transit (`transit_id`) I took the unweighted means of `epoch` (TCB) and `g_mag` over its CCD observations; G_i is this mean. Its uncertainty is $\sigma_i = \sqrt{(1.0857 \max \sigma_F/F)^2 + 0.03^2}$, with the maximum taken over the CCDs and a floor for calibration errors. JPL Horizons (observer code 500@-139479, Gaia), queried at the transit epochs, gave the heliocentric distance r_i and the distance from Gaia Δ_i (au) and the phase angle α_i (deg). Epochs t_i (days) were corrected for light time, $t_i \to t_i - \Delta_i/c$. A new *season* starts wherever consecutive transits are more than 60 d apart; no minimum number of transits per season is imposed.

METHOD

For a trial frequency f (cycles per day) the reduced magnitude is modelled as

$$G_i - 5\log_{10}(r_i\Delta_i) = c_{s(i)} + \beta\,\alpha_i + \sum_{k=1}^{2}\left[a_k\cos(2\pi k f t_i) + b_k\sin(2\pi k f t_i)\right], \tag{1}$$

where $s(i)$ is the season of transit i. All coefficients are free and linear: one offset c_s per season, which absorbs changes of aspect between seasons, a common phase slope β (mag deg^{-1}), and four Fourier amplitudes. They are solved by least squares with weights σ_i^{-2}, giving $\chi^2(f)$. No correction for the motion of the phase-angle bisector is applied, so periods are synodic. Before the search, the model without Fourier terms is fitted, transits whose residual (in mag) exceeds 5×1.4826 times the median absolute deviation of the residuals are removed, and fit and cut are repeated once.

$\chi^2(f)$ is evaluated on a uniform grid from 0.24 to 12 cycles per day with a step of $0.1/T$, where T is the time span. The grid minimum f_1 gives $P_1 = 1/f_1$. Because most lightcurves are double-peaked and can be fitted by the second harmonic at half the period, I adopt $P = 2P_1$ when $\chi^2(f_1/2)$, computed exactly, is below $1.5\,\chi^2(f_1)$ and $2P_1 \le 100$ h; otherwise $P = P_1$. The factor 1.5 maximises the number of validation periods within 3% of the reference ($|P - P_{\rm ref}|/P_{\rm ref} < 0.03$; half or double counts as disagreement) over factors 1.00–3.00 in steps of 0.05. The amplitude is the peak-to-peak range of the Fourier terms fitted at $1/P$.

The confidence score is $S = \ln(\chi_0^2/\chi_1^2)\,(\chi_2^2/\chi_1^2 - 1)$, with χ_0^2 the fit without Fourier terms, $\chi_1^2 = \chi^2(f_1)$ whether or not the period is doubled, and χ_2^2 the lowest grid value outside $\pm 5/T$ of f_1, $f_1/2$ and $2f_1$ (of $f_1/4$ instead of $2f_1$ when $\chi^2(f_1/2) < 1.25\,\chi^2(f_1)$, a fixed setting of the search code). Sorting the validation results by decreasing S, the

threshold $S \geq 0.1604$ is the score of the last asteroid at which the cumulative agreement fraction is still at least 80% (from the tenth asteroid on), using periods doubled with the factor 1.5. Code and the list of VizieR tables are archived with the data (Tropynin 2026a).

VALIDATION

With the doubling factor chosen on one half of the validation set (alternate asteroids in the ranked list) and tested on the other, and vice versa, 203 of 300 periods agree. Above the threshold:

- validation set: 172 of 214 agree (in-sample, because this set fixed the threshold);
- target-set asteroids solved from the same Gaia data with a different method by Ďurech & Hanuš (2023): 81 of 97 agree;
- the other confident target-set asteroids with a rotation period in Cellino et al. (2024), MacLennan et al. (2026) or McNeill et al. (2023): 15 of 25 agree.

Out of sample, 96 of 122 (79%) agree. Binned by the period found here, 165 of 203 (2–10 h), 78 of 88 (10–30 h), 15 of 17 (30–50 h) and 10 of 28 (50–100 h) agree, counting all three groups.

THE 34 CANDIDATES

Of the 298 target-set asteroids, 165 pass the threshold. I removed the 97 solved by Ďurech & Hanuš (2023), the 25 above, and 35277, whose only published period, 23.61 h (Liberato et al. 2024), comes from astrometric wobble but is close to ours (23.88 h). None of the remaining 42 has a rotation period, in October 2026, in JPL SBDB, LCDB V4.0, DAMIT (Ďurech et al. 2010), the SsODNet ssoBFT table (Berthier et al. 2023), 179 VizieR tables (5.1 million rows), or the Minor Planet Bulletin volumes 50–53, except 11857, for which Ďurech et al. (2022) give 19.72 h from ATLAS photometry; I dropped it. A second analysis of all of Gaia DR3 (Tropynin 2026b) adds a correction for the phase-angle bisector, a calibrated probability that a period is correct, and searches to 1000 h. I dropped the seven candidates where its best period differs by more than 3%: four where it is twice ours and three where it is a slow solution. The remaining 34 (Table 1) are main-belt asteroids with 58–91 transits in 4–7 seasons; the best period of the second analysis is within 0.2% of ours for all of them, with probability above 0.8 for 18. Formal period uncertainties are not computed. I offer the list as targets for dense-lightcurve confirmation.

This work used data from the ESA mission Gaia, processed by the Gaia DPAC; JPL Horizons and the JPL Small-Body Database; and VizieR (CDS, Strasbourg). The analysis was designed, coded and checked with the help of an AI assistant (Claude, Anthropic) under the author's direction.

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Table 1. The 34 candidate rotation periods.

Number	P (h)	Amp.	N	Q	Number	P (h)	Amp.	N	Q
5897	6.0729	0.39	67	A	44204	4.7663	0.72	65	A
8751	3.2402	0.25	70	A	45726	7.9396	0.33	69	B
11621	33.7373	0.60	59	A	47917	4.7480	0.38	80	A
11818	2.7435	0.17	66	B	50588	56.9297[a]	0.55	65	A
13363	8.9320	0.26	68	A	52653	3.7362	0.14	65	–
16916	3.8301	0.15	69	B	55335	5.0076	0.14	61	–
17337	4.9700	0.21	65	B	56994	14.7458	0.39	64	A
22427	5.4857	0.46	66	A	63007	4.4476	0.33	65	A
24359	16.1525	0.22	62	A	64303	8.8781	0.19	63	B
24711	6.4557	0.75	61	B	65213	3.8679	0.34	73	A
30068	8.3416	0.27	72	B	72885	3.0007	0.32	62	B
31611	5.4767	0.38	62	A	84508	3.0394	0.23	69	A
32778	66.9411[a]	0.56	63	A	87249	4.4550	0.19	70	B
35653	30.0890	0.13	66	–	88237	52.2814[a]	0.10	58	–
40172	5.1295	0.54	74	A	91539	5.6743	0.42	91	A
42087	81.9684[a]	0.22	66	B	97578	2.8696	0.41	86	A
44017	3.0915	0.20	65	B	105992	8.0425	0.19	58	B

NOTE—P: adopted period after the doubling rule. Amp.: peak-to-peak amplitude of the Fourier terms (mag). N: transits used after outlier removal. Periods are given beyond the grid step of about $0.1P^2/T$. Q: tier in the second analysis (Tropynin 2026b), A for probability ≥ 0.8, B for $0.384 \leq p < 0.8$, – below. [a]Above 50 h; the true period may be twice the value given.